MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 810

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

See figure 1

Mounting position: Vertical with base down, or in horizontal position with pins 1 and 2 in vertical plane.

Weight: 8 ounces (226.8 grams) nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec	Ib	Ic	Pp	Pi
Unit:	MHz	V	Vdc	Vdc	mAdc	mAdc	W	W
Maximum:	15	10.5	2,000	-500	250	70	125	500
Minimum:	---	9.5	---	---	---	---	---	---
TEST CONDITIONS:	---	10.0 Vdc (Note 2)	2,000	-35	---	---	---	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	4.2	4.8	Adc
1266	Total grid current	-	Ec/Ib = 75 mAdc; t = 300 seconds	0.65	II	Ic	0	-10	µAdc
1256	Electrode current (anode)	-		0.65	II	Ib	39.0	71.0	mAdc
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Eb = 250 Vdc; Ec = +5 Vdc; Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1256	Electrode current (anode)	-	Eb = 200 Vdc; Ec = 100 Vdc	---	---	Ib	390	690	mAdc
1266	Grid current (positive)	-	Eb = 200 Vdc; Ec = 100 Vdc	---	---	Ic	67	148	mAdc
1316	Amplification factor	-	Ec/Ib = 55 mAdc	---	---	Mu	32.0	40.0	---
1236	Power oscillation (1)	-	Rg = 5,000 ohms; Ic = 45 to 70 mAdc; Ib = 250 mAdc; F = 15 MHz	---	---	Po	340	---	W
1231	Emission by oscillation	1	Power oscillation (1)	---	---	Ef	---	9.0	Vdc
1236	Internal insulation	-	F = 15 MHz	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-	No voltages applied; no external shield; base shell floating	---	---	Cgp Cgf Cpf	4.2 6.5 9.0	5.6 10.2 15.0	pF pF pF
	Quality conformance inspection, part 3								
---	Life test	-	Group C; Pp = 125 W; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Emission by oscillation	-		---	---	Ef	---	9.5	Vdc
1236	Power oscillation (2)	3	Power oscillation (1); F = 30 MHz	---	---	Po	310	---	W

NOTES:

1. This test is to be the first test performed at the conclusion of the holding period.

(A) 2. When using dc on the filament, then the negative filament terminal shall be ground and serve as the reference point for the other dc electrode voltages.

(A) 3. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

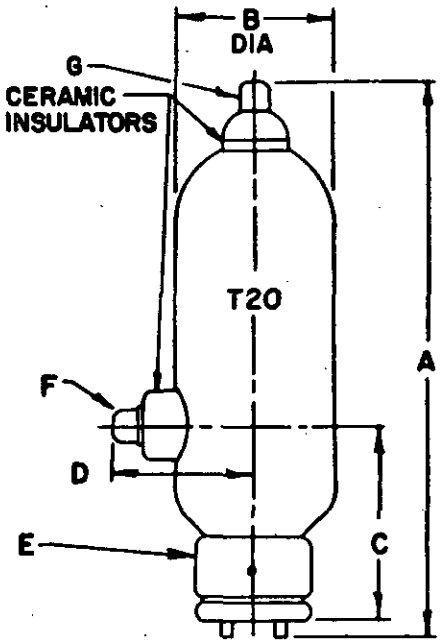
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2961)

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Pin connections	
Pin No.	*Element*
1	nc
2	f
3	nc
4	f
Top cap	a
Side cap	g

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	8.210 (208.53)	9.060 (230.12)
B		2.563 (65.10)
C		3.187 (80.95)
Quality conformance inspection, part 3 (See note 3)		
D		2.250 (57.15)
E	Base: A4-29 (EIA)	
F	Grid cap: C1-5 (EIA)	
G	Anode cap: C1-14 (EIA)	

FIGURE 1. Outline drawing of electron tube type 810.

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